



Name: Gargi Sukhatankar
Roll No: B027

SAP ID: 60019220125
Branch: CSE(ICB)

EXPERIMENT 11

Aim: To study and implement Storage as a Service using GCP.

Screenshots:

Create bucket

The screenshot shows the Google Cloud Console interface for creating a new bucket. The left sidebar contains navigation links for Overview, Buckets, Monitoring, Storage Insights, and Settings. The main content area is titled 'Create a bucket' and includes a 'Get Started' section with a text input field for a bucket name (example: 'example', 'example_bucket-1', or 'example.com') and a 'CONTINUE' button. Below this are sections for 'Choose where to store your data' (Location: us, Location type: Multi-region), 'Choose a storage class for your data' (Default storage class: Standard), 'Choose how to control access to objects' (Public access prevention: On, Access control: Uniform), and 'Choose how to protect object data' (Soft delete policy: Default, Object versioning: Disabled, Bucket retention policy: Disabled, Object retention: Disabled, Encryption type: Google-managed). A 'Good to know' section on the right provides information on location pricing and current configuration (Multi-region / Standard).

Item	Cost
us (multiple regions in United States)	\$0.026 per GB-month
With default replication	\$0.020 per GB written

ESTIMATE YOUR MONTHLY COST

Select region



Name: Gargi Sukhatankar
Roll No: B027

SAP ID: 60019220125
Branch: CSE(ICB)

[←](#) Create a bucket

☒

Get Started
| Name: cclty2025_b2

☐

Choose where to store your data
This choice defines the geographic placement of your data and affects cost, performance, and availability. Cannot be changed later. [Learn more](#) [↗](#)
Location type

☒ **Multi-region**
Highest availability across largest area

us (multiple regions in United States) ▼

☐ **Add cross-bucket replication via Storage Transfer Service**
As data is added or changed, replicate it to another bucket, enabling you to store a copy that follows different bucket settings - e.g., region, storage class, etc.
[Learn more](#) [↗](#)

☐ **Dual-region**
High availability and low latency across 2 regions

☐ **Region**
Lowest latency within a single region

CONTINUE

☐

Choose a storage class for your data
| Default storage class: Standard

☐

Choose how to control access to objects
| Public access prevention: On
Access control: Uniform

☐

Choose how to protect object data

Cross bucket replication



Name: Gargi Sukhatankar
Roll No: B027

SAP ID: 60019220125
Branch: CSE(ICB)

☒ **Add cross-bucket replication via Storage Transfer Service**

As data is added or changed, replicate it to another bucket, enabling you to store a copy that follows different bucket settings - e.g., region, storage class, etc.

[Learn more](#)

Bucket *

 cclty2025_b1

BROWSE

Ex. mybucket

Replication settings [CONFIGURE](#)

Unless you configure custom settings below, all changes in your bucket will be replicated to the destination bucket, with replicated data following the settings of the destination.

Set object storage class to	Use destination bucket's storage class
Include objects with prefixes	Include all objects
Exclude objects with prefixes	—
Service account type	Storage Transfer Service service agent

Choose storage class



Name: Gargi Sukhatankar
Roll No: B027

SAP ID: 60019220125
Branch: CSE(ICB)

- **Choose a storage class for your data**

A storage class sets costs for storage, retrieval, and operations, with minimal differences in uptime. Choose if you want objects to be managed automatically or specify a default storage class based on how long you plan to store your data and your workload or use case. [Learn more](#)



Autoclass is not supported for buckets with hierarchical namespaces.



Autoclass

Automatically transitions each object to Standard or Nearline class based on object-level activity, to optimize for cost and latency. Recommended if usage frequency may be unpredictable. Can be changed to a default class at any time. [Pricing details](#)



Set a default class

Applies to all objects in your bucket unless you manually modify the class per object or set object lifecycle rules. Best when your usage is highly predictable.



Standard

Best for short-term storage and frequently accessed data



Nearline

Best for backups and data accessed less than once a month



Coldline

Best for disaster recovery and data accessed less than once a quarter



Archive

Best for long-term digital preservation of data accessed less than once a year

[CONTINUE](#)



Name: Gargi Sukhatankar
Roll No: B027

SAP ID: 60019220125
Branch: CSE(ICB)

- **Choose how to protect object data**

Your data is always protected with Cloud Storage but you can also choose from these additional data protection options to add extra layers of security.

Data protection



Object versioning and retention are not yet available for buckets with hierarchical namespaces.



Soft delete policy (For data recovery)

When enabled, this bucket and its objects will be kept for a specified period after they're deleted and can be restored during this time. [Learn more](#)



Use default retention duration

All buckets have a 7 day soft delete duration by default, unless this default has been customized by your organization administrator.



Set custom retention duration

Specify how long this bucket and its objects should be retained after they're deleted. Setting a "0" duration disables soft delete, meaning any deleted objects will be permanently deleted.



Object versioning (For version control)

For restoring deleted or overwritten objects. To minimize the cost of storing versions, we recommend limiting the number of noncurrent versions per object and scheduling them to expire after a number of days. [Learn more](#)



Retention (For compliance)

For preventing the deletion or modification of the bucket's objects for a specified period of time.

Storage bucket is created



**SHRI VILEPARLE KELAVANI MANDAL'S
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING**
(Autonomous College Affiliated to the University of Mumbai)
NAAC ACCREDITED with "A" GRADE (CGPA : 3.18)



Name: Gargi Sukhatankar
Roll No: B027

SAP ID: 60019220125
Branch: CSE(ICB)

← Bucket details

📁 cclty2025_b2

Location

us-central1 (Iowa)

Storage class

Standard

Public access

Not public

Protection

Soft Delete

Hierarchical namespace

Enabled

OBJECTS

CONFIGURATION

PERMISSIONS

PROTECTION

LIFECYCLE

OBSERVABILITY

NEW

INVENTORY REPORTS

OPERATIONS

Folder browser

📁 cclty2025_b2

Buckets > cclty2025_b2

CREATE FOLDER

UPLOAD

TRANSFER DATA

OTHER SERVICES

Filter by name prefix only

Filter

Filter objects and folders

Name

Size

Type

Created

Storage class

Last modified

Public access

Version history

Encryption

Configuration of the bucket

← Bucket details

OBJECTS

CONFIGURATION

PERMISSIONS

PROTECTION

LIFECYCLE

OBSERVABILITY

NEW

INVENTORY REPORTS

OPERATIONS

Overview

Created

March 25, 2025 at 10:53:18 AM GMT+5

Updated

March 25, 2025 at 10:53:18 AM GMT+5

Hierarchical namespace

Enabled

Location type

Region

Location

us-central1 (Iowa)

Replication

—

Cross-bucket replication

Not enabled

Default storage class

Standard

Requester Pays

OFF

Tags

None

Labels

None

Cloud Console URL

https://console.cloud.google.com/storage/browser/cclty2025_b2

gsutil URI

gs://cclty2025_b2

Permissions

Access control

Uniform

Public access prevention

Enabled via bucket setting

Public access status

Not public

Protection

Soft delete policy

7 days

Object versioning

Off

Bucket retention policy

None

Object retention

Disabled

Encryption type

Google-managed

Object lifecycle

Lifecycle rules

None

Anywhere Cache

Cache

Not configured

Compute zones

—



Name: Gargi Sukhatankar
Roll No: B027

SAP ID: 60019220125
Branch: CSE(ICB)

Permission setting for the bucket

OBJECTS

CONFIGURATION

PERMISSIONS

PROTECTION

LIFECYCLE

OBSERVABILITY

NEW

Soft delete policy (for data recovery)

When enabled, deleted objects will be kept for a specified period after they're deleted and can be restored during this time. [Learn more](#)

EDIT

DISABLE

Retention period

7 days

Effective date

March 25, 2025 at 10:53:18 AM GMT+5

Object versioning (for data recovery)

With object versioning on, you can restore objects that have been overwritten or deleted. Live and noncurrent versions are stored in the same bucket and storage class by default. To reduce costs, limit the number of versions by adding a lifecycle rule. [Learn more](#)

OBJECT VERSIONING OFF

Retention (for compliance)

Prevents the deletion or modification of the objects in this bucket for a specified period of time after they're uploaded. [Learn more](#)

Bucket retention policy ?

Best to guarantee compliance for all objects in this bucket for a uniform retention period. The optional step of setting the retention mode to "locked" ensures that no one can remove or shorten the retention period.

SET RETENTION POLICY

Object retention ?

Best to guarantee compliance at the individual object level to meet specific retention needs. The bucket must have this option enabled before objects in it can have their own retention configuration.

Lifecycle of the bucket



SHRI VILEPARLE KELAVANI MANDAL'S
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC ACCREDITED with "A" GRADE (CGPA : 3.18)



Name: Gargi Sukhatankar
Roll No: B027

SAP ID: 60019220125
Branch: CSE(ICB)

OBJECTS CONFIGURATION PERMISSIONS PROTECTION **LIFECYCLE** OBSERVABILITY **NEW** INVENTORY REPORTS OPERATIONS

After you add or edit a rule, it may take up to 24 hours to take effect.

Lifecycle rules let you apply actions to a bucket's objects when certain conditions are met – for example, switching objects to colder storage classes when they reach or pass a certain age. [Learn more](#)

If an object meets the conditions for multiple rules:

- Deletion takes precedence over a change in storage class.
- Changing objects to colder storage classes takes precedence over changing to warmer ones (ex. objects will switch to the Archive storage class instead of Coldline if there are rules for both).

Rules [ADD A RULE](#) [DELETE ALL](#)

Action	Object condition	Works with
--------	------------------	------------

You haven't added any lifecycle rules to this bucket.

Folder is created in the bucket and file is uploaded

Cloud Storage

Overview

Buckets

Monitoring

Storage Insights

Settings

Bucket details

cclty2025_b2

Location

us-central1 (Iowa)

Storage class

Standard

Public access

Not public

Protection

Soft Delete

Hierarchical namespace

Enabled

OBJECTS

CONFIGURATION

PERMISSIONS

PROTECTION

LIFECYCLE

OBSERVABILITY

INVENTORY REPORTS

OPERATIONS

Folder browser

cclty2025_b2

B2/

Buckets > cclty2025_b2 > B2

CREATE FOLDER

UPLOAD

TRANSFER DATA

OTHER SERVICES

Filter by name prefix only

Filter objects and folders

Name	Size	Type
Identity and Access Management...	1.3 MB	application/vnd.openxmlformats-officedocument.wordproc...

1 file successfully uploaded



SHRI VILEPARLE KELAVANI MANDAL'S
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC ACCREDITED with "A" GRADE (CGPA : 3.18)



Name: Gargi Sukhatankar
Roll No: B027

SAP ID: 60019220125
Branch: CSE(ICB)

Data can be accessed by the following ways:

Cloud Console URL

https://console.cloud.google.com/storage/browser/cclty2025_b2

gsutil URI

gs://cclty2025_b2

Conclusion:

Thus, we successfully implemented Storage as a Service using GCP.